

Phase 01 - Heat up + Calibration

Explanation and Operating Logic - Unified GUI

Release: 0.9.36 **Build:** 2026.08.11-r15

Operator and developer explanation of the current Phase 01 unified GUI, control flow, calibration logic, and safety behavior.

1. Role in the synthesis workflow

- Heat the CK-1 evaporator under Keysight current control.
- Monitor CK-1 and Sample QMB, chamber pressure, oven PID, CK-1 temperature and pyrometer channels.
- Enter calibration after the physical shutter is opened and confirmed.
- Calculate the final CK-1 relative thickness / Sample relative thickness ratio used by Phase 03.

2. Unified launcher and GUI

Unified launcher: START_NPG_CHAMBER.bat opens the main launcher. Each phase starts as a separate child process and owns its serial ports and hardware cleanup.

Phase-local shutdown: Abort / Safe Stop closes only the active phase after its safe hardware action. The unified launcher remains open and can be used again after the child process releases its ports.

Launcher close protection: while a phase is active, the launcher itself refuses to force-kill it. Use the phase GUI Abort / Safe Stop or let the phase finish normally.

GUI area	Purpose
Live plots	QMB thickness/rate, pressure, temperature channels, Keysight current and voltage.
Editable controller	Temperature/rate targets, feedback mode, ramp settings and manual current override.
Operator controls	Open shutter, Close shutter, Abort / safe stop, Finish phase.
Status	Current phase, targets, measured values and last accepted action.

3. Heating and shutter transition

Shutter readiness is intentionally minimal and auditable.

Condition	Requirement
CK-1 temperature	At or above the active target/guide.
Average CK-1 rate	At or above the requested rate target.
Hidden equilibrium diagnostics	None. Controller slope/trend/cascade quantities do not gate shutter opening.

When both physical process conditions are true, the state advances from HEATING_UP to WAIT_SHUTTER_OPEN. The operator must still physically open the shutter and confirm it in the GUI.

4. Calibration and normal automatic completion

- Open Shutter confirmation stores CK-1 and Sample QMB baselines.
- Calibration uses relative thickness from those baselines.

- At the sample calibration target, the GUI asks for shutter close.
- After shutter-close confirmation, the normal controller enters RAMP_DOWN and reduces Keysight current to 0 A before OUTPUT OFF.
- The final ratio is saved for Phase 03.

5. Abort versus Finish Phase

Action	Electrical result	Process result
Abort / safe stop	Immediate CURR 0.000, then OUTP OFF.	No normal ramp-down. Save what is available, close only Phase 01, launcher stays open.
Finish phase	Controlled Phase 01 RAMP_DOWN to 0 A, then OUTP OFF.	Treat as a normal manual completion and close only Phase 01 when ramp-down finishes.

Safety ordering: Abort sends the Keysight zero/OFF command before snapshot or summary file I/O, so logging cannot delay removal of evaporator power.

6. Safety layers kept active

- Maximum automatic current cap: 0.660 A by default.
- Fixed software hard current stop: 0.680 A.
- Independent absolute CK-1 watchdog maximum: 255 C by default.
- Voltage/OCP/OVP and unexpected-output-off handling remain independent from the PID/Compound controller.
- QMB plausibility filtering protects feedback from obviously corrupted readings.

7. Data and operator checklist

- Run data are stored under Data Samples/Heat up + Calibration Data/.
- Check live channels before opening the shutter.
- Use Finish phase for a normal manual end; use Abort only when you want immediate electrical shutdown.
- After a normal calibration, verify and retain the saved thickness ratio.

Operational note: Software state is not a substitute for checking the real chamber hardware. Verify shutters, pressure, PID, Keysight output and any manually operated hardware before continuing.